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Laboratory Experiment # 003: Information Theory

This notebook explores the fundamental limits of communication systems as defined by Information Theory. We will analyze and visualize:

1. **Information Measure (Entropy):** The theoretical limit of data compression.
2. **Nyquist Capacity:** The maximum rate for a noiseless channel.
3. **Shannon-Hartley Capacity:** The maximum rate for a noisy channel.
4. **Eb/No Analysis:** System performance in terms of energy and noise.

Key Libraries:

- `numpy` : For numerical calculations and logarithmic functions.
- `matplotlib` : For visualizing relationships (Entropy curves, Capacity limits).

```
In [6]: import numpy as np
import matplotlib.pyplot as plt

# Configure plot style for readability
plt.rcParams['figure.figsize'] = [10, 6]
plt.rcParams['axes.grid'] = True
plt.rcParams['font.size'] = 12

print("Libraries loaded successfully.")
```

Libraries loaded successfully.

1. Information Measure & Entropy

Information is measured in **bits**. The average information of a source is called **Entropy (H)**.

For a binary source (two possible messages) with probability p and $(1 - p)$, the entropy is given by:

$$H = -p \log_2(p) - (1 - p) \log_2(1 - p)$$

Analysis: The code below calculates and plots the Entropy of a binary source as probability p varies from 0 to 1. Note where the maximum entropy occurs.

```
In [2]: def plot_binary_entropy():
        """
        Plots the Entropy H(p) for a binary source (Bernoulli trial).
        """
```

```

# 1. Define Probability Range (avoid 0 and 1 to prevent log(0) error)
p = np.linspace(0.001, 0.999, 100)

# 2. Calculate Entropy Formula
#  $H = -p \log_2(p) - (1-p) \log_2(1-p)$ 
h = -p * np.log2(p) - (1 - p) * np.log2(1 - p)

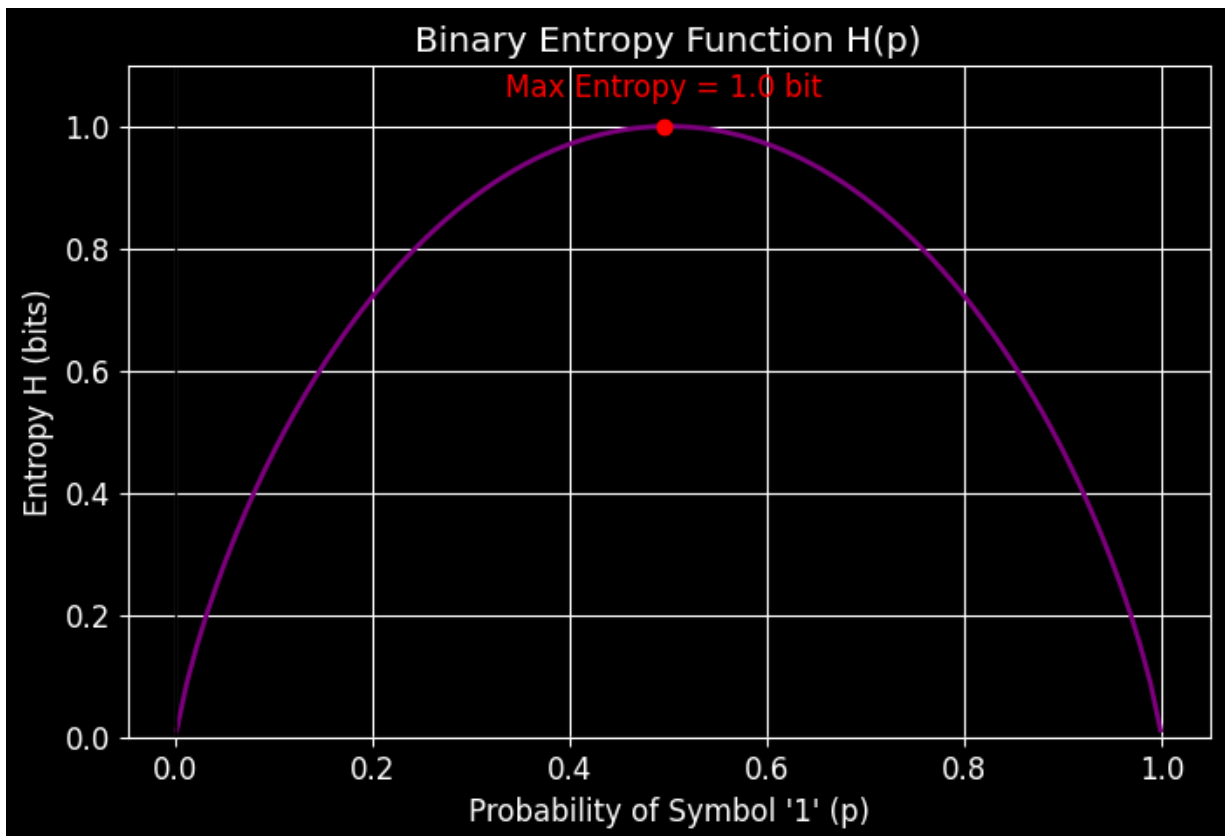
# 3. Plotting
plt.figure(figsize=(8, 5))
plt.plot(p, h, linewidth=2, color='purple')

# Annotate Max Entropy
max_h_idx = np.argmax(h)
plt.plot(p[max_h_idx], h[max_h_idx], 'ro') # Mark the peak
plt.text(p[max_h_idx], h[max_h_idx] + 0.05, f'Max Entropy = {h[max_h_idx]:.1f}',
         ha='center', color='red')

plt.title("Binary Entropy Function H(p)")
plt.xlabel("Probability of Symbol '1' (p)")
plt.ylabel("Entropy H (bits)")
plt.axhline(0, color='black', lw=1)
plt.axvline(0, color='black', lw=1)
plt.ylim(0, 1.1)
plt.show()

# --- Exercise ---
# Run the function to visualize how uncertainty relates to probability.
plot_binary_entropy()

```



2. Nyquist Capacity (Noiseless Channel)

For an ideal **noiseless** channel, the capacity is limited by bandwidth (B) and the number of signaling levels (M).

Formula:

$$C = 2B \log_2 M$$

Analysis: We will plot the Channel Capacity (C) against Bandwidth (B) for different modulation levels (e.g., $M=2$ for Binary, $M=16$ for 16-QAM).

```
In [3]: def plot_nyquist_capacity():
        """
        Plots Capacity vs Bandwidth for different M-ary encoding levels.
        """
        # 1. Define Bandwidth Range (e.g., 0 to 10 kHz)
        bandwidth = np.linspace(0, 10000, 100)

        # 2. Define M-levels to compare
        m_levels = [2, 4, 8, 16]
        colors = ['blue', 'green', 'orange', 'red']

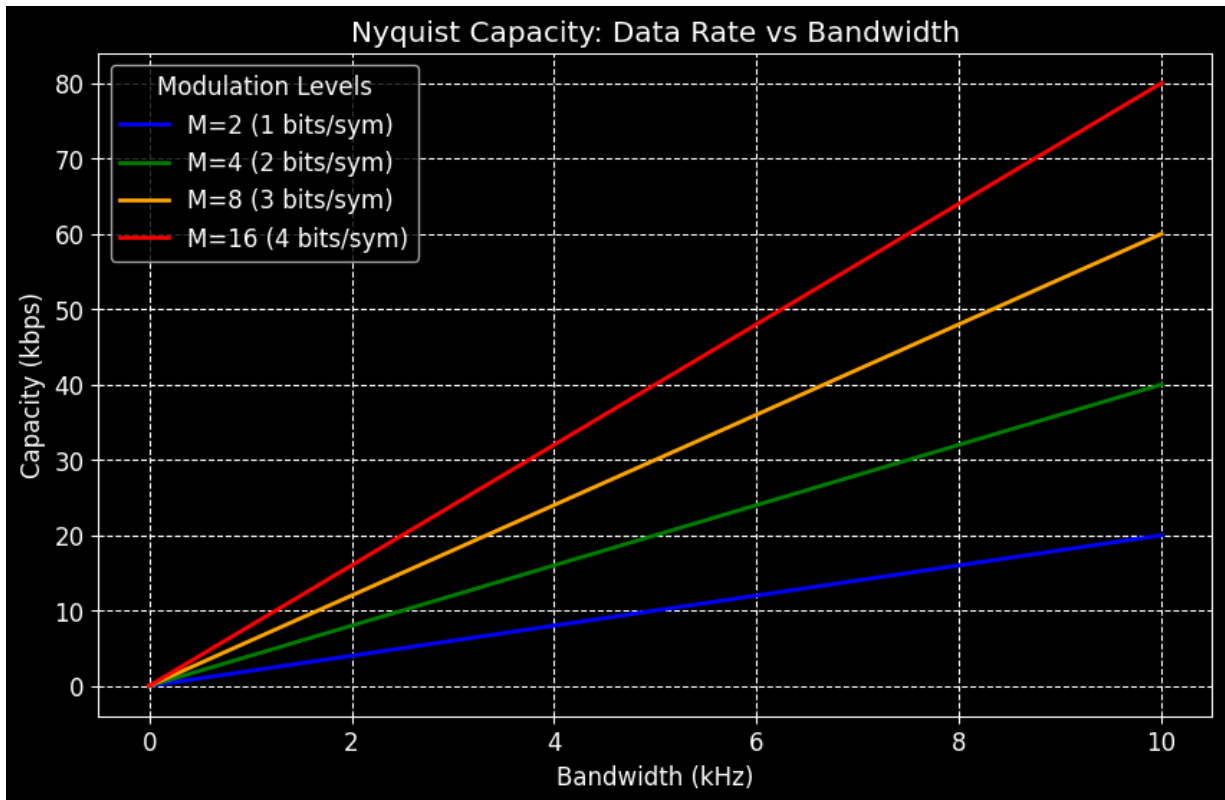
        # 3. Plotting
        plt.figure(figsize=(10, 6))

        for m, color in zip(m_levels, colors):
            # C = 2 * B * Log2(M)
            capacity = 2 * bandwidth * np.log2(m)

            # Convert to kbps for cleaner y-axis
            plt.plot(bandwidth/1000, capacity/1000, label=f'M={m} ({int(np.log2(m))} bi

        plt.title("Nyquist Capacity: Data Rate vs Bandwidth")
        plt.xlabel("Bandwidth (kHz)")
        plt.ylabel("Capacity (kbps)")
        plt.legend(title="Modulation Levels")
        plt.grid(True, which='both', linestyle='--')
        plt.show()

        # --- Exercise ---
        # Observe how increasing M (bits per symbol) increases the slope of the line.
        plot_nyquist_capacity()
```



3. Shannon-Hartley Law (Noisy Channel)

In the real world, noise limits performance. Shannon's Law defines the theoretical maximum data rate (C) for a noisy channel.

Formula:

$$C = B \log_2(1 + S/N)$$

Analysis: We will plot the maximum achievable Data Rate vs. Signal-to-Noise Ratio (SNR) in dB. This is the fundamental "speed limit" of the internet.

```
In [4]: def plot_shannon_limit(bandwidth=3000):
        """
        Plots Capacity vs SNR (dB) for a fixed bandwidth.
        Default Bandwidth: 3000 Hz (Voice channel).
        """
        # 1. Define SNR Range in dB (e.g., 0 dB to 60 dB)
        snr_db = np.linspace(0, 60, 200)

        # 2. Convert dB to Linear Power Ratio
        # SNR_linear = 10^(SNR_db / 10)
        snr_linear = 10 ** (snr_db / 10)

        # 3. Calculate Shannon Capacity
        # C = B * Log2(1 + S/N)
        capacity = bandwidth * np.log2(1 + snr_linear)

        # 4. Plotting
```

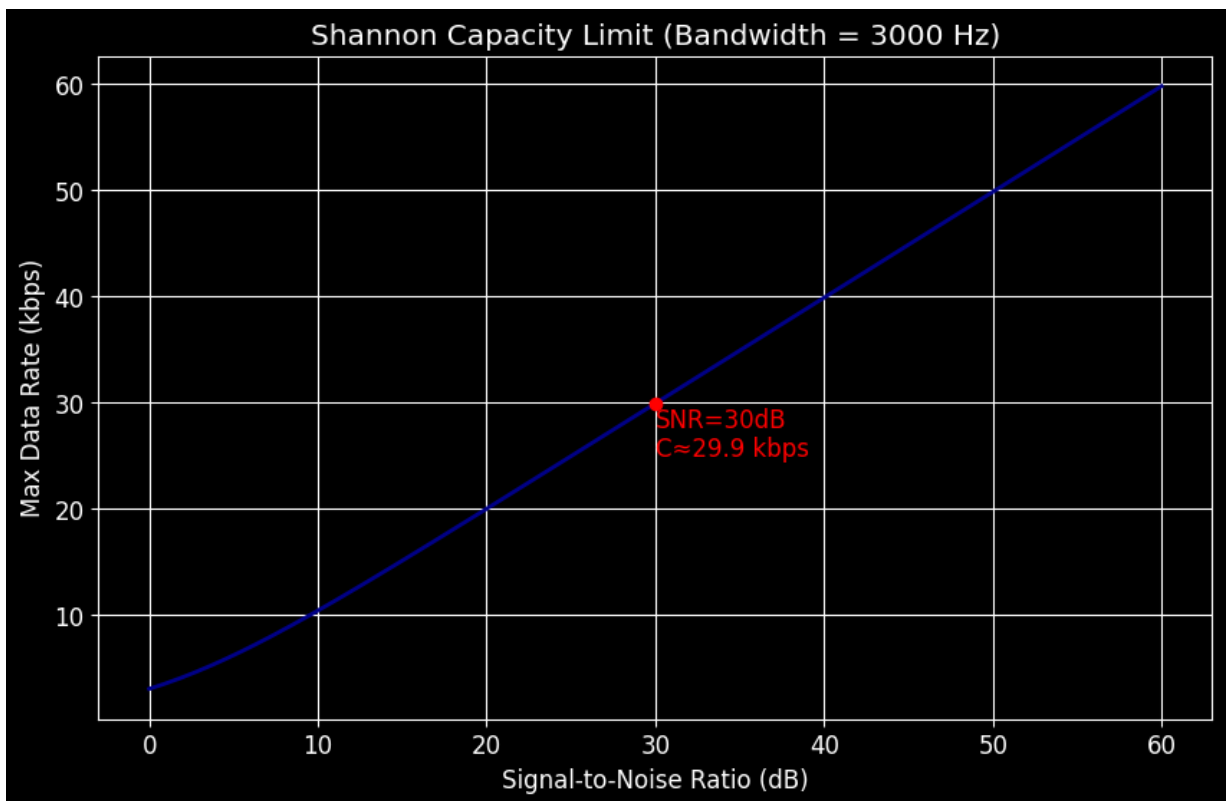
```
plt.figure(figsize=(10, 6))
plt.plot(snr_db, capacity/1000, color='darkblue', linewidth=2)

plt.title(f"Shannon Capacity Limit (Bandwidth = {bandwidth} Hz)")
plt.xlabel("Signal-to-Noise Ratio (dB)")
plt.ylabel("Max Data Rate (kbps)")

# Add marker for a typical scenario (e.g., 30 dB)
target_snr = 30
target_cap = bandwidth * np.log2(1 + 10**(target_snr/10)) / 1000
plt.plot(target_snr, target_cap, 'ro')
plt.text(target_snr, target_cap - 5, f'SNR=30dB\nC≈{target_cap:.1f} kbps', color='red')

plt.grid(True, which="both", ls="--")
plt.show()

# --- Exercise ---
# Run the function to see the Logarithmic relationship between Power (SNR) and Speed
plot_shannon_limit(bandwidth=3000)
```



4. Eb/No vs. Spectral Efficiency

Often we want to compare different systems independent of bandwidth. We use **Spectral Efficiency** (C/B) vs **Energy per bit** (E_b/N_o).

The Shannon Limit in terms of E_b/N_o :

$$\frac{C}{B} = \log_2 \left(1 + \frac{C}{B} \cdot \frac{E_b}{N_o} \right)$$

This plot shows the "Shannon Region" (achievable) vs. the "Unachievable Region". The absolute minimum E_b/N_o required for reliable communication is **-1.6 dB**.

```
In [5]: def plot_ebno_limit():
        """
        Plots Spectral Efficiency (C/B) vs Required Eb/No (dB).
        This is the famous 'waterfall' limit curve.
        """

        # 1. Define Spectral Efficiency range (bits/s/Hz)
        rho = np.linspace(0.1, 10, 100) # C/B

        # 2. Calculate required Eb/No (linear)
        # Formula derived from Shannon: Eb/No = (2^rho - 1) / rho
        ebno_linear = (2**rho - 1) / rho

        # 3. Convert to dB
        ebno_db = 10 * np.log10(ebno_linear)

        # 4. Plotting
        plt.figure(figsize=(10, 6))
        plt.plot(ebno_db, rho, linewidth=2, color='teal', label='Shannon Limit')

        # Shade the regions
        plt.fill_betweenx(rho, ebno_db, 30, color='green', alpha=0.1, label='Achievable')
        plt.fill_betweenx(rho, -10, ebno_db, color='red', alpha=0.1, label='Unachievable')

        # Mark the -1.59 dB Limit
        plt.axvline(-1.59, color='red', linestyle='--', alpha=0.7)
        plt.text(-1.4, 8, 'Shannon Limit\n(-1.6 dB)', color='red')

        plt.title("Spectral Efficiency vs. Energy Efficiency (Eb/No)")
        plt.xlabel("Eb/No (dB)")
        plt.ylabel("Spectral Efficiency (bits/s/Hz)")
        plt.xlim(-5, 25)
        plt.ylim(0, 10)
        plt.legend()
        plt.grid(True)
        plt.show()

        # --- Exercise ---
        # This plot separates what is physically possible from what is impossible.
        plot_ebno_limit()
```

